EEEN30160 – Biomedical Signal Processing

Assignment 1

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Problem 1:

Code:

```
function plotsinusoid2(f,a,startT,endT,step)
    %plot a complex sinusoid of frequency, 'f' and amplitude, 'a',
with 'startT'
    %and 'endT' giving the bouns of the time and 'step' giving the
step size
    t = (startT:step:endT); % create array of timesf from start to
end in the specified step size
    compx = a*exp(2*pi()*f*t*i); % create a complex number using the
specified inputs
    Rex = real(compx);%get the real component of compx
    Imx = imag(compx);%get the imaginary component of compx
    Ax = (t*0)+a;% create an amplitude array that is the same size
as t
    tiledlayout(2,1)% create a 2 graph tiled layout to plot the
angle and x
    nexttile %go to the first tile
    plot(t,Rex, 'r--') % plot the real part of x in red dashed line
    hold on % keep the real part of x
    plot(t,Imx, 'b-.'); % plot the imaginart part of x in a blue dot
and dashed line
    plot(t,Ax,'g-') % plot the amplitude of x
    xlabel('time') % label x axis as time
    ylabel('amplitude')% label y axis as amplitude
    ylim ([-a-1 a+1]) % add 1 to the y bounds, as to increase the
amount of graph shown
    legend('Re(x)', 'Im(x)', 'Amp(x)')%add a legend
    title('PlotSinusoid2 - Amplitude')%add a title
    nexttile % next tile
    plot(t,(360*f*t))% Plot the angle vs time
    xlabel('time')%label x
    ylabel('Angle')%label y
    title('PlotSinusoid2 - Angle')% title

    return
end
```

Output

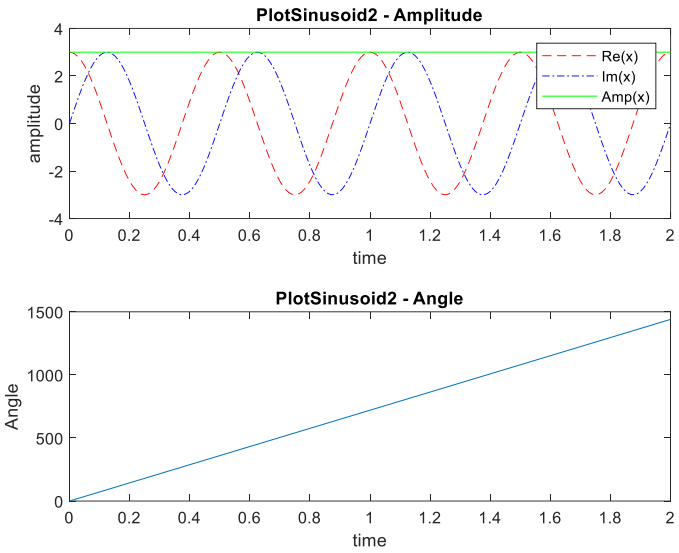

Figure 1

Discussion

The real and imaginary parts of the complex number are separated by a quarter step. The angle graph could be changed to reset to 360 each iteration using the modulo function, mod(a,x) , but this was not deemed necessary, when this was done, the graph looked like this:

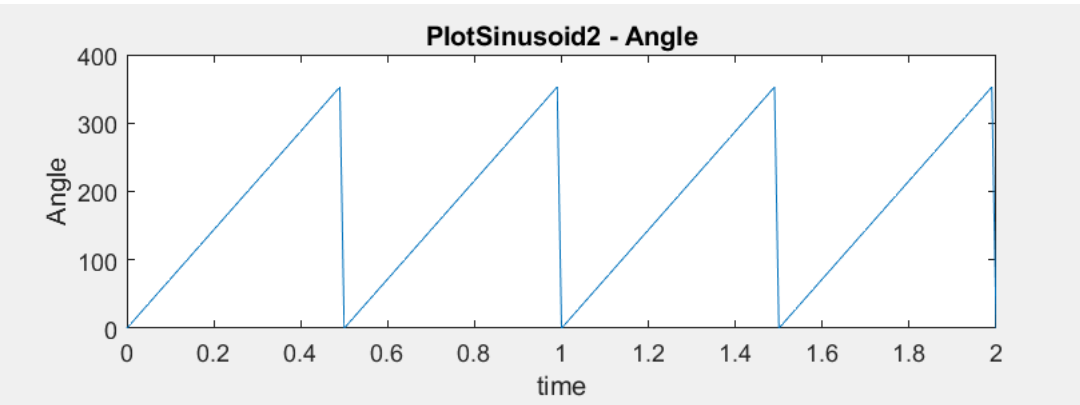

Figure 2

Problem 2:

Code:

```
a = load('C:\Users\Oisin\Documents\MATLAB\Signal processing\EvokedPotential.mat'); % load file as 'a'
b = a.EvokedPotential;%let 'b' equal the evoked potential part of 'a'
l = size(b); % let 'l' equal the size of 'b'
t = [-100:650]; % let t range from -100 to 650 ms
plot(t, b)% plot 'b' against 't', each 1 on the x axis is equal to 1 ms
xlabel('time - ms')% label the x axis as time in ms
```

Outputs:

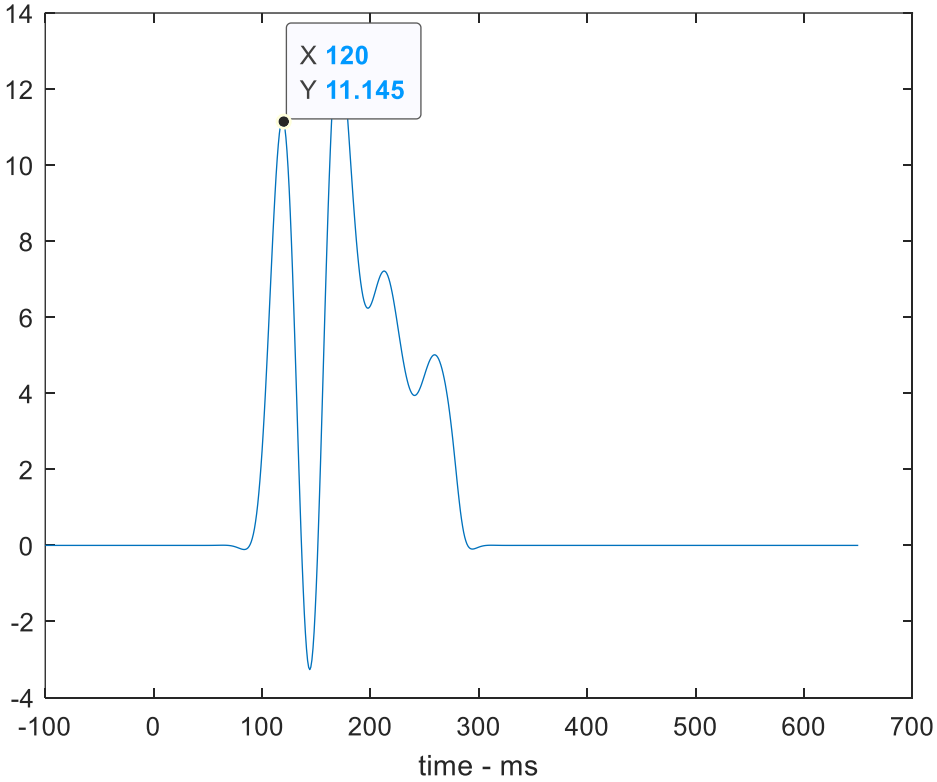

Figure 3

Discussion:

The file was read into MATLAB and then plotted from the range of -100ms to 650ms, when this was done, the peak was found to be at 120ms, meaning that the subject was neurologically healthy.

Problem 3:

Code:

```
function ans = rmsfunct(n) % function to calculate the rms of a
siganl
    x = n.^2*sqrt(2);% let x be the square of each element in the
array n, times root 2
    N = size(n);% let N be the size of x
    ans =sqrt(1/N(2)*sum(x.^2)); % ans equal ro the rms, done with
the given function in the problem
end

n = [-728:0.5:1275]; % let n be an array from -728 to 1275 in steps
of 0.5
rmsfunct(n)% calculate the rms of n
```

Output:

```
ans =


   8.4514e+05
```

Discussion

The root mean square of this signal was 85140, when compared to the built in MATLAB function rms(), this number was confirmed

Problem 4:

Code:

```
struct = load('C:\Users\Oisin\Documents\MATLAB\Signal
processing\EMG.mat');% load in EMG signal
vect = struct.EMG; %let 'vect' be the EMG signal
avect = abs(vect); %get absolute value of vect
plot(avect) %plot avect

lb = 4664;% lower bound observed from graph
ub = 11759;% upper bound observed from graph
avs =avect(lb:ub); %set the signal component of 'avect'

avnb = avect(1:lb); % set the lower bound of the noise component of
'avect'
avna= avect(ub:16001); % set the upper bound of the noise component
of 'avect'
avn = [avnb; avna];% add them both together to get the total noise
of 'avect'
ravn = rms(avn);% get rms of noise
ravs = rms(avs);% get rms of signal


SNR = 20*log(ravs/ravn) % use the rms values to get the SNR
```

Output:

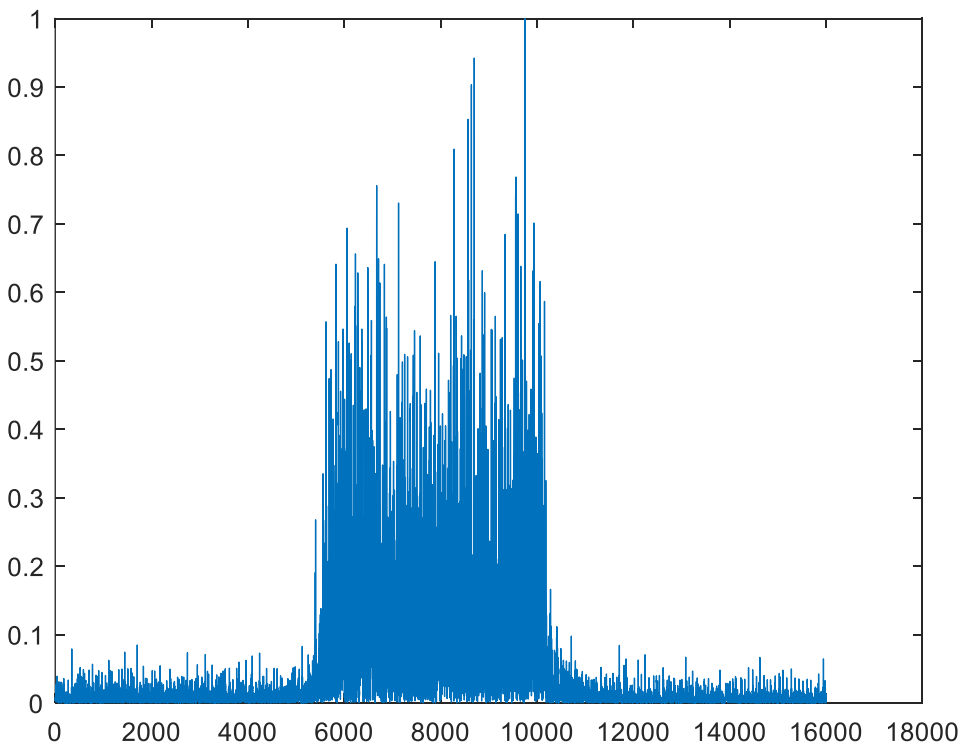

Figure 4

```
SNR =

   51.0796
```

Discussion:

The signal frequency was Hz, and the noise frequency was Hz. The signal noise ratio was 51.0796.

Problem 5

Code:

```
struct = load('C:\Users\Oisin\Documents\MATLAB\Signal
processing\vep.mat'); % read in the vep file
sigs = struct.vep;% get the signals from struct
sizes = size(sigs);% get the dimensions of sigs
Diff_Min_Max = max(sigs)- min(sigs); % find the difference between
the min and max values of signal
avgdiff = mean(Diff_Min_Max);% find the mean difference of the
signals
filteredvep = mean((sigs.'));% filtered vep is the mean of all the
signals, .' transposes the matrix to ensure its the right dimensions
plot(sigs, '--')% plot all the signals in red dashed lines
hold on% hold all the signals
plot(filteredvep, 'K')%plot the filtered vep in black
ylabel('mV')% label the y axis as mV
maxv = max(filteredvep); % find the max of the filtered vep
minv = min(filteredvep); %find the min of the filtered vep
tmax = find(filteredvep == maxv);% find the time that the max occurs
tmin = find(filteredvep ==minv);% find the time that the min occurs
tdiff = abs(tmax - tmin);% find the difference in times
diffavg = maxv- minv;% find the average difference in the largest
and smallest value
var20 = (std(sigs(1:20, :))).^2; % find the varience of the first 20
time steps of the signas
avgvar20 = mean(var20); % find the mean of these variences
varavg20 = (std(filteredvep(1:20)))^2; % vind the varience of the
first 20 time steps of the filtered signal
```

Output:

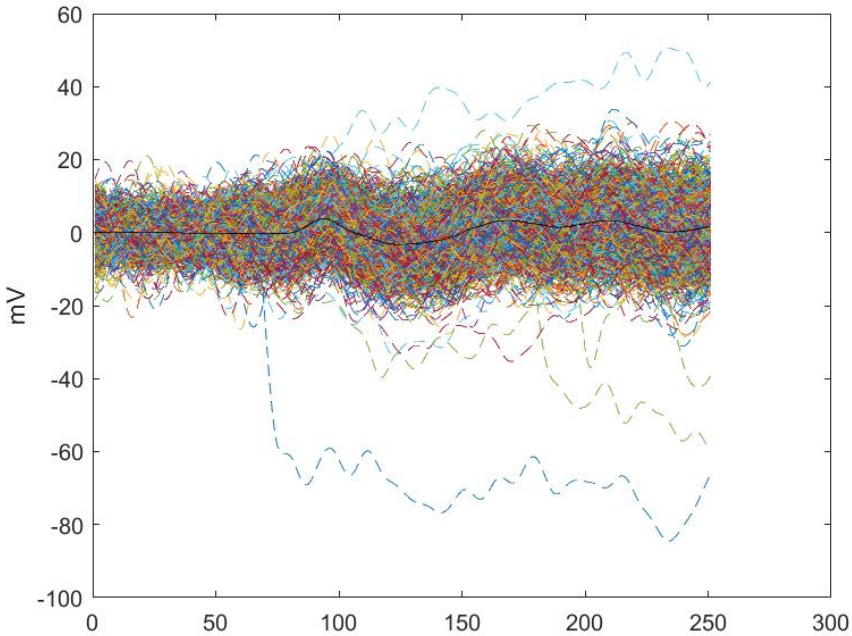

Figure 5

Discussion:

A

Figure 5 has all the signals plotted together, all shown in different colour dotted lines

B

The Diff_Min_Max was a 1 x 1263 array, where it was found by subtracting the max and min arrays from each other.

C

The average value for Diff_Min_Max was found to be 24.230395622150300

D

The filtered vep, or the average epoch is shown in Figure 5 in black.

E

The max and min points of the vep occurred at 94 samples and 124 samples, with a time between occurrences of 30 samples.

F

Because the average difference across all the trials looks at the extreme values, and the filtered signal has softened the large peaks of them, leading to a much smaller difference between the extremes.

G

The variance of the different epochs for the first 20 frames was a 1263 x 1 array, with the average value of 10.439121218095861

H

The variance of the first 20 samples of the filtered vep was 0.007396304600416, this was a lot lower than the average variance, as the extreme outliers that would have a large effect on the data were removed in the vep, leading to a much lower varience